Candidate Name: ______________________________ Class: 07S_________

JURONG JUNIOR COLLEGE

J2 Preliminary Exam 2008

H2 Biology 9747/3

THURSDAY 21 AUGUST 2008 1 HOUR 30 MIN

INSTRUCTIONS TO CANDIDATES:

Write your name and class in the spaces at the top of this page and on any separate answer paper used.

Answer **all** questions.

At the end of the examination
1. fasten any separate answer paper used securely to the question paper.

EXAMINER'S USE ONLY	
1	
2	
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Total	

INFORMATION FOR CANDIDATES:

The intended number of marks is given in brackets [ ] at the end of each question or part question.

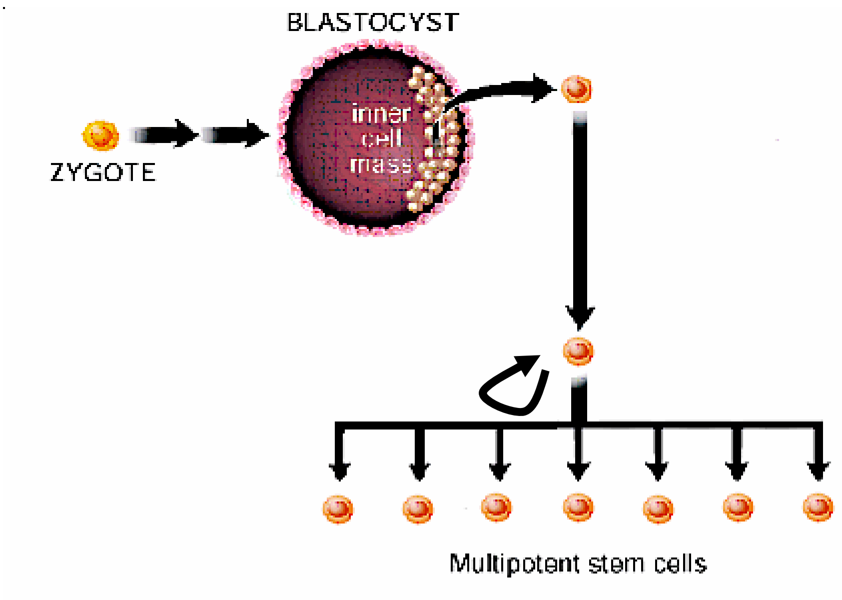
This Paper consists of **8** printed pages.

[Turn over]

Section A

Answer **all** questions.

1. The figure below shows how a one-celled zygote undergoes various cellular divisions to give rise to other cell types.



(a) Describe two unique features of stem cells that could be seen from the figure above. [2]

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(b) Describe the developmental potential of the one-celled zygote and the inner cell mass of the blastocyst. [4]

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(c) Multipotent stem cells can form specialized cell types of the tissue in which they reside. Give an example of a multipotent stem cell and the specialized cell types that it can formed. [2]

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(d) Adult stem cells are multipotent stem cells found among differentiated cells in a tissue or organ. What are the primary roles of these adult stem cells and describe one appropriate unique feature of this type of stem cells that is not mentioned in (a). [2]

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Parkinson’s disease is a disorder of the central nervous system involving the brain and spinal cord. Parkinson’s disease occurs when certain nerve cells in a part of the brain become impaired. A chemical substance called dopamine is made by certain cells in the brain. Dopamine carries messages that tell the body when and how to move, allowing for smooth, coordinated functions of the body’s muscles. Parkinson’s disease occurs when these cells either die or become damaged. When this happens there isn’t enough dopamine to carry the messages and movement becomes very difficult.

Treatment methods involved implantation of stem cells that transform into dopamine-producing cells into the brains of Parkinson’s disease patients. In one method, one man’s own neural stem cells were extracted from his brain and stimulated to differentiate into dopamine-secreting cells in vitro. These cells are subsequently transplanted into the left side of the brain.

Another treatment method involved cultivating stem cells from another tissue so that they become dopamine-producing cells and injecting them into target sites of the patients’ brain that need dopamine. These transplantations of cells have enabled patients to remain active.

(e) Describe how the stem cells are differentiated into dopamine-secreting cells *in vitro*. [3]

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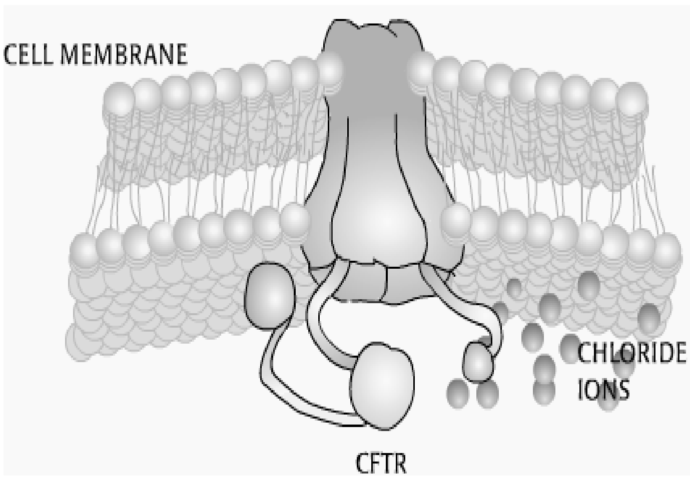
(f) “One treatment method involved cultivating stem cells from another tissue so that they become dopamine-producing cells…”

State the property of stem cells that allows for such transformation to occur. [1]

[Total: 14 marks]

2. The first British trial of a possible gene therapy for cystic fibrosis (CF) began in 1993. Nine adult male patients with advanced lung disease were chosen. Six patients received gene therapy and three acted as a control.

   The purpose of the trial was to see whether functioning ion channels could be manufactured by cells which have had the normal allele of the CF gene inserted into them. DNA coding for the normal allele was cloned and put into small spheres of phospholipids bilayers (liposomes) which were introduced into lung epithelial cells.



   (a) Describe the gene mutation and mode of inheritance of cystic fibrosis. [4]

(b) Discuss why the liposome strategy is considered an example of gene therapy? [2]

(c) Describe how a liposome introduces DNA into mammalian cells. [2]

(d) Compare and contrast retroviral and adenoviral vectors used in gene therapy. [4]

(e) Explain how the expression of the mutant cystic fibrosis alleles leads to the occurrence of bacterial infections in lungs. [3]

(f) Suggest why male patients were chosen for the British trial. [1]

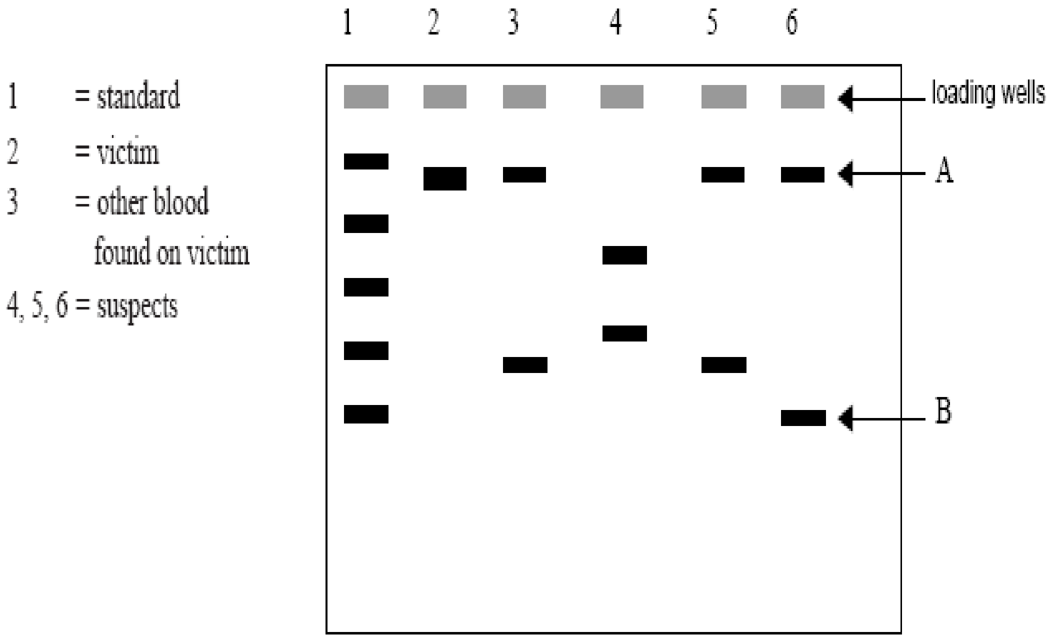
[Total: 16 marks]

3. There were three suspects in an assault case. A forensic scientist found blood, other than the victim's, at the site. DNA was extracted from five blood samples:
   - The victim
   - The blood at the assault site (not the victim’s)
   - The three suspects

   Polymerase Chain Reaction (PCR) was carried out on the extracted DNA from the five blood samples.

   One of the regions used in the forensic analysis was a short tandem repeat (STR) sequence of 4 bases, called **HUMTHO1**. This sequence, located on chromosome 11, has many alleles which differ from each other by the number of times the sequence **AATG** is repeated. It was this region of chromosome 11 which was amplified using PCR.

   The amplified samples were loaded onto a gel and electrophoresis was performed to separate the fragments of DNA. A diagram of the gel is shown in the figure below.



(a) Why is there only one band in lane 2 but two bands in lanes 3, 4, 5 and 6? [2]

(b) How many different alleles at the **HUMTHO1** locus are represented on the gel in individuals 2, 3, 4, 5, 6? [1]

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(c) Which piece of DNA, A or B, has the greater number of the 4 base repeat sequence? Explain why. [2]

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(d) Which of the suspects appears to have committed the assault? Explain your choice. [2]

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(e) Explain briefly the basis behind Restriction Fragment Length Polymorphisms (RFLP). [2]

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(f) State another application of RFLP analysis besides genetic fingerprinting. [1]

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(g) Explain why genes especially housekeeping genes are seldom used as markers for genetic fingerprinting. [2]

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The Human Genome Project (HGP) was one of the great feats of exploration in history, an inward voyage of discovery of the genome of members of our species, *Homo sapiens.* The HGP was completed in April 2003. Now we can, for the first time, read nature's complete genetic blueprint for building a human being.

(h) Suggest how information from HGP can help scientists understand the role of genes in complex diseases like cancer. [3]

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[Total: 15 marks]

Section B

Answer the following question on the separate answer paper provided.
Your answer must be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections (a), (b) etc., as indicated in the question.

4. With reference to the analysis of DNA molecules,

   a. Describe and explain the use of the Polymerase Chain Reaction (PCR). [10]
   b. Describe the process and explain the use of Gel Electrophoresis and Nucleic Acid Hybridisation. [10]